

VAL256

The French fleet that opened the network in 1996 — and spent eighteen months in a depot having its brain replaced while the line it built ran without it.

BR WENHU LINE 文湖線			
BUILDER Matra Transport, with GEC Alsthom ^[1]	FAMILY VAL	OPERATOR Taipei Rapid Transit Corporation	LINE Wenhu
BUILT 1990—1993 ^[1]	IN SERVICE 28 March 1996 — ^[6]	DEPOT Muzha and Neihu ^[7]	



A VAL256 set in service.

Stub page. This entry is an outline. Figures marked *TBC* still need to be checked against a primary source before publication.

Formation



4 CARS · 2 UNITS OF 2 · — PERMANENTLY COUPLED — COUPLING | CAB

A train delivered in the early 1990s is still running the line, and almost nothing tells it what to do any more.

The VAL256 sets were built by Matra with GEC Alsthom between 1990 and 1993^[1] and entered service with the Muzha Line [\[/rail/lines/wenhu-line/\]](/rail/lines/wenhu-line/) on 28 March 1996^[6]. The 256 is the car width in centimetres — 2,560 mm — and it is the wider of the two standard VAL profiles, the other being the 2,060 mm VAL206 used at Lille^[4].

The re-signalling

The conversion did not precede the Neihu [\[/rail/stations/br19/\]](/rail/stations/br19/) extension — it was triggered by it, and the sequence is the story:

DATE	EVENT
4 July 2009	The Neihu extension opens. The line switches to CITYFLO 650, the Matra system is cut off after thirteen years, and the VAL256 fleet comes out of service that day to begin conversion ^[1] ^[5]
26 December 2010	Converted VAL256 units return to passenger service, running alongside the Innovia fleet ^[1]

So for nearly eighteen months the whole 24-station line ran on the Bombardier fleet alone, while the trains that had opened the railway sat in a depot having their control system replaced. Why the line was re-signalled at all — and why by a competitor of the system's owner — is told on The Matra dispute [\[/rail/history/matra-dispute/\]](/rail/history/matra-dispute/).

The original equipment was a fixed-block ATC/ATO system derived from the MAGGALY technology on Lyon Metro line D^[2]; what replaced it is a moving-block communications-based system.

Design

VAL vehicles run on rubber tyres over concrete running surfaces, and are guided from the **sides**, not the centre: pairs of horizontal wheels bear outward against lateral guide bars — H-section steel, set roughly 200 mm above the running surface on either side of the vehicle — and those same bars carry the 750 V supply, which is why a VAL guideway needs no separate conductor rail^[3]^[4].

VAL's own successor went the other way. The NeoVal generation Siemens announced in 2006 *is* centrally guided, on a single rail similar to Translohr's^[4]. Central guidance is VAL's future, not its past, and Taipei has the past.

Cars are built as permanently coupled married pairs; two pairs make a four-car train in normal service, so the 51 pairs form 25 trains with one pair spare^[8].

Fleet history

Ordered for the Muzha Line and in service from its opening on 28 March 1996, originally working BR01 through BR12 only. Since December 2010 the fleet has run the full route alongside the Innovia APM 256 (C370) [\[/rail/rolling-stock/innovia-apm-256-c370/\]](/rail/rolling-stock/innovia-apm-256-c370/).

Delivery dates by unit, and the scope of the re-signalling work on each set, are still to be confirmed.

Depot allocation

Maintained at Muzha Depot [\[/rail/depots/muzha-depot/\]](/rail/depots/muzha-depot/), beside BR01, which also stables Innovia sets ^[7].

Corrections

- **This page previously said the fleet was re-signalled ahead of the 2009 extension.** That inverts the sequence: the conversion began the day the extension opened and the fleet was out of service until December 2010 ^[1].
- **This page previously said VAL vehicles are guided by horizontal wheels bearing on a central rail.** They are side-guided ^[3] ^[4]. The likely origin of the error is the NeoVal succession — the *later* generation is centrally guided, and the two were run together here.

Specifications

Car width	2.56 m ^[1]
Car length	13.78 m ^[1]
Car height	3.53 m ^[1]
Cars per train	4 ^[1]
Empty weight	TBC t
Design maximum speed	80 km/h ^[1]
Operating speed	70 km/h ^[1]
Seated capacity	20 per car (AW2) ^[1]
Standing capacity	94 per car (AW2) ^[1]
Train capacity	440 ^[9]
Doors per car	4 ^{2 per side} ^[1]
Power supply	750 V DC ^[1]
Traction	GEC Alsthom chopper DC ^[1]
Motors per car	2 ^[1]
Braking	Regenerative and disc ^[1]
Guidance	Side guide bars ^[3]
Signalling	CITYFLO 650 since 2010 ^[1]
Automation	GoA4 (UTO) ^[5]
Fleet size	102 cars ^[1]
Married pairs	51 ^[1]
Tyre replacement interval	TBC

References

2 of 9 sources on this page are primary — published by the operator, the builder or government.

- ¹ Taipei Metro VAL256 electric multiple unit [https://zh.wikipedia.org/zh-tw/台北捷運VAL256型電聯車]

台北捷運VAL256型電聯車

SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-06

The fullest specification found for this fleet. The article carries no reference section, so every figure taken from it is one source deep — corroborated below by the English article, which is independently worded.

2 Taipei Metro VAL256 [https://en.wikipedia.org/wiki/Taipei_Metro_VAL256]

SECONDARY English Wikipedia accessed 2026-08-06

Agrees with the Chinese article on every dimension, capacity, speed and date. Adds that the original signalling derived from the MAGGALY system on Lyon Metro line D.

3 Lille VAL — project profile [https://www.railway-technology.com/projects/lille_val/]

SECONDARY Railway Technology accessed 2026-08-06

Describes the guideway directly — running surfaces, lateral H-section guide bars about 200 mm above the running surface, and those bars doubling as the 750 V supply. Trade press, not a manufacturer drawing; a Siemens or Matra engineering document would settle it properly.

4 Véhicule Automatique Léger [https://en.wikipedia.org/wiki/V%C3%A9hicule_Automatique_L%C3%A9ger]

SECONDARY English Wikipedia accessed 2026-08-06

Independent corroboration of side guidance — "pairs of horizontal tires to provide lateral guidance", with power "collected by shoes from the guidebars". Also records that the later NeoVal moved to a single central rail, which is the most likely origin of the central-rail error this page used to carry.

5 Wenhua line [<https://zh.wikipedia.org/zh-tw/文湖線>]

文湖線

SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-06

6 Wenshan—Neihu Line [<https://www.dorts.gov.taipei/cp.aspx?n=DBAC040496EFAB94>]

文山內湖線

PRIMARY Taipei City Government, Department of Rapid Transit Systems (DORTS)

accessed 2026-08-06

archived [<https://web.archive.org/web/20260209210832/https://www.dorts.gov.taipei/cp.aspx?n=DBAC040496EFAB94>]

7 Muzha Depot [<https://zh.wikipedia.org/zh-tw/木柵機廠>]

木柵機廠

SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-06

8 Taipei Metro turns space magician: after the Bombardier fleet, the Wenhua line's Matra trains get an interior optimisation [https://www.metro.taipei/News_Content.aspx?n=30CCEFD2A45592BF&sms=72544237BBE4C5F6&s=86F98235BC105C44]

臺北捷運化身空間魔術師！繼龐巴迪列車改善後 文湖線馬特拉列車空間優化

PRIMARY Taipei Rapid Transit Corporation (news release) accessed 2026-08-06

archived [https://web.archive.org/web/20250723182554/https://www.metro.taipei/News_Content.aspx?n=30CCEFD2A45592BF&sms=72544237BBE4C5F6&s=86F98235BC105C44]

The operator's own announcement of the seat-removal programme, and the source for this fleet forming 25 four-car trains.

9 **The whole Wenhua line to be renewed — new trains gangwayed across four cars, capacity up 30% to 574** [<https://www.ettoday.net/news/20260507/3162180.htm>]

文湖線將全線更新 新列車採「四節貫穿」、載客量增30%至574人

SECONDARY ETtoday 新聞雲 , 7 May 2026 accessed 2026-08-06

Gives 440 as the current Matra train capacity, as the comparison for the replacement programme. Treat as provisional — it is a figure quoted in a council exchange reported in news, not a published specification, and it does not reconcile exactly with 114 per car.

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Station names, codes and sequence from TDX Transport Data eXchange, Ministry of Transportation and Communications, operator TRTC. Government open data. Retrieved 2026-08-05; source last updated 2020-01-31. Provenance.